

Exercise Sheet #1

Basics

Exercise 1: Coulomb Force

Two point particles are separated by a distance of 60 cm and carry a total charge of $200\text{ }\mu\text{C}$. Determine the charge of each of the two particles if they

- a) repel each other with a force of 80 N,
- b) attract each other with a force of 80 N.

Exercise 2: Ohm's Law

A resistor of $15\text{ }\Omega$ is connected to a voltage of 30 V. What are the current and the conductance?

Exercise 3: Resistance

Given are two copper wires of equal mass with circular cross-sections; wire A is twice as long as wire B. How do the resistances of the wires compare (neglecting temperature effects)?

- a) $R_A = 8R_B$,
- b) $R_A = 4R_B$
- c) $R_A = 2R_B$,
- d) $R_A = R_B$.

Exercise 4: Resistivity

Suppose we have a copper wire with a resistance of $2\text{ }\Omega$. We want to replace the copper wire with an aluminum wire of the same length and resistance. By what factor must the diameter of the wire be increased?

Exercise 5: Temperature Dependence of Resistance

At what temperature does a copper wire have a resistance that is 10% greater than at $20\text{ }^\circ\text{C}$?

Exercise 6: Energy

A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

Exercise 7: Power

Find the hot resistance of a light bulb rated 60 W, 120 V.